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Optimization of energy use efficiency and environmental assessment in soybean and peanut farming using the imperialist competitive algorithm

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ABSTRACT

This study proposes using input-output analysis to improve energy use efficiency and assess environmental indicators in soybean and peanut farming. The research findings show that soybean production consumes 43170.20 MJ ha^{-1} of total energy and yields 40417.28 MJ ha^{-1} of output energy, while peanut production requires 28677.36 MJ ha^{-1} of total energy and produces 28677.36 MJ ha^{-1} of output energy. The LCA results indicate that on-farm emissions are associated with production inputs for both soybean and peanut systems, with diesel fuel and chemical fertilizers being the main contributors. Soybean farming shows higher levels of diesel fuel pollutants due to reduced usage, while peanuts have a higher prevalence of contaminants associated with diesel fuel. Using ICA to calculate optimized energy consumption results in 36672.77 MJ ha^{-1} for soybean production and 23485.45 MJ ha^{-1} for peanut production, leading to energy savings of 6075.33 MJ ha^{-1} and 5191.91 MJ ha^{-1}, respectively. Additionally, the environmental impact category quantities for ICA analysis of soybean and peanut production reveal the resources per ton for soybean production to be 111.25 USD2013 and for peanut production to be 45.43 USD2013 in terms of energy use. The use of ICA demonstrates potential in maximizing energy efficiency and evaluating environmental indicators in soybean and peanut farming. By optimizing resource allocation and decision-making processes, ICA can help farmers reduce energy consumption and minimize environmental impact, ultimately leading to more sustainable and eco-friendly farming practices that benefit both the environment and the agricultural industry. Further research and implementation of ICA in farming practices have the potential to significantly improve energy efficiency and environmental sustainability in the soybean and peanut farming industry.

1. Introduction

Farming soybeans and peanuts is crucial for producing essential food products and boosting the economy. However, these agricultural practices also have a notable environmental impact and demand substantial energy (Altieri et al., 2012). Maximizing energy efficiency and assessing environmental indicators in soybean and peanut farming is crucial for sustainable agricultural practices. Maximizing energy efficiency involves finding ways to reduce the amount of energy used to perform a specific task or operate a system (Mondal et al., 2021; Simpeh et al., 2022). Utilizing renewable energy sources allows businesses and individuals to decrease their dependence on fossil fuels and decrease their total energy usage. Furthermore, adopting energy-efficient methods in transportation, such as carpooling, using public transportation, and investing in fuel-efficient vehicles, can also help to maximize energy efficiency. In general, maximizing energy efficiency is crucial for cutting energy consumption, reducing energy expenses, and lessening environmental harm (Savelli and Morstyn, 2021; Nabavi-Pelesaraei et al., 2023). It requires a combination of technological advancements, behavioral changes, and policy initiatives to achieve significant improvements in energy efficiency across various sectors of society. Assessing environmental indicators involves evaluating various aspects of the environment to understand its current state and potential impacts (Ruparathna et al., 2016). This assessment can include analyzing air and water quality, biodiversity, land use, and other factors that contribute to the overall health of the environment. Environmental indicators are

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used to measure and monitor changes in the environment over time, and they provide valuable information for decision-making and policy development. Assessing these indicators involves collecting data, analyzing trends, and identifying potential areas of concern or improvement (Singh et al., 2012). Environmental assessments can be conducted at various scales, from local to global, and can help identify areas where intervention is needed to protect the environment and promote sustainability. By assessing environmental indicators, stakeholders can better understand the state of the environment and make informed decisions to mitigate environmental degradation and promote conservation efforts (Lyche Solheim et al., 2019; Nabavi-Pelesaraei and Damgaard, 2023). The Imperialist Competitive Algorithm (ICA) is a metaheuristic optimization algorithm inspired by the concept of imperialism in history. In the ICA, candidate solutions are represented as colonies, and the algorithm simulates the process of imperialist expansion and competition for resources (Sherinov and Ünveren, 2018). The algorithm consists of two main processes: the colonization process, where colonies are developed and improved, and the competition process, where the most dominant colonies (imperialists) compete for resources and influence. During the colonization process, colonies improve their solutions through local search and collaboration with other colonies (Liu et al., 2013). The competition process involves the selection of the most dominant colonies (imperialists) based on their fitness values, and the weaker colonies are assimilated into the dominant ones. The ICA aims to find the optimal solution to a given optimization problem by iteratively improving the candidate solutions through the processes of colonization and competition (Hadidi et al., 2013). It has been successfully applied to various optimization problems in fields such as engineering, economics, and finance. Overall, the ICA is a powerful and versatile optimization algorithm that draws inspiration from historical imperialism to solve complex optimization problems. The ICA is a metaheuristic optimization algorithm inspired by the socio-political behavior of imperialist countries (Kalashnikov et al., 2018). It has been widely used in various fields, including agriculture, to solve complex optimization problems. In the context of soybean and peanut farming, the ICA can be employed to optimize energy consumption and assess environmental indicators to minimize the environmental impact of farming activities (Tang et al., 2021). Taherzadeh-Shalmaei et al. (2023) focused on examining energy consumption and environmental emissions within the mushroom production industry. They accomplished this by conducting an energy audit analysis and a life cycle assessment (LCA). The energy flow analysis revealed that the total input energy for mushroom production was 1022537.82 MJ/m^2, while the total output energy was only 11125.94 MJ/m^2. This led to an energy use efficiency rate of 0.01, indicating significant energy imbalance and inefficiency. Nabavi-Pelesaraei et al. (2021a,b) the research aims to evaluate the exergoenvironmental efficiency of utilizing solar technologies for sunflower oil production in Iran. The energy analysis indicates that the production of 1 ton of sunflower oil requires approximately 180, 354 MJ of energy and yields about 39,400 MJ. Alluvione et al. (2011) examined energy flows in a wheat-maize-soybean-maize rotation under three different cropping systems: low-input integrated farming (LI), integrated farming following European Regulations (IFS), and conventional farming (CONV). The results indicate that balancing nitrogen fertilization with actual crop requirements and implementing minimum tillage are the most effective methods for reducing energy inputs, contributing 64.7% and 11.2% respectively to the overall reduction.

The use of the ICA in soybean and peanut farming is unique in its ability to maximize energy efficiency and assess environmental indicators. This nature-inspired optimization technique can be applied to various agricultural processes, including planting and harvesting schedules, irrigation and fertilization practices, and pesticide and herbicide usage. By utilizing this algorithm, farmers can make more informed decisions about their farming practices, ultimately leading to more sustainable and environmentally friendly farming methods. The study aims to optimize energy usage and evaluate the environmental impact of soybean and peanut cultivation, ultimately contributing to the development of more environmentally friendly and energy-efficient agricultural systems.

2. Materials and methods

2.1. Case study regine

This study took place in Parsabad, Moghan, a city in the Ardabil province of northwest Iran. Positioned on the southern bank of the Aras River and to the west of the Caspian Sea, Parsabad lies between 39′ and 12° to 39′ and 42° north latitude, and 47′ and 10° to 48′ and 21° east longitude from the Greenwich meridian (Ministry of Jihad-e-Agriculture of Iran, 2021). This region is a significant hub for agriculture in Iran. This study involved the calculation of a sample size of 150 farmers using Equation (1) to assess their awareness and attitudes towards input consumption and crop production. The sample size was determined based on similar studies and the formula for cross-sectional studies, with a significance level of p = 0.05 and an accuracy of d = 0.07 (Cochran, 1977). Data was collected using a standard questionnaire, which was structured based on previous research with minor modifications. The questionnaire consisted of three main parts: personal information and background, irrigation methods, and awareness of water conservation practices.

$$n = \frac{\frac{z^2pq}{d^2}}{1 + \frac{1}{N}\left(\frac{z^2pq}{d^2} - 1\right)} \tag{1}$$

2.2. Energy use analysis

The process of energy use analysis involves assessing and understanding the utilization of energy within a system, building, or organization. This includes gathering and studying data on energy consumption, identifying energy usage patterns and trends, and evaluating the effectiveness of energy systems and equipment (Ghasemi-Mobtaker et al., 2020). The goal of energy use analysis is to identify opportunities for improvement and to develop strategies for reducing energy consumption, increasing energy efficiency, and lowering energy costs. This may involve conducting energy audits, using energy monitoring and management systems, and implementing energy-saving technologies and practices. Key steps in energy use analysis may include (Altieri et al., 2012; Nabavi-Pelesaraei et al., 2021a,b).

1. Data collection: Gathering information on energy consumption, including electricity, natural gas, and other fuels, as well as data on building or system characteristics, operating schedules, and occupancy patterns.
2. Data analysis: Analyzing energy use data to identify trends, patterns, and anomalies, and to understand the factors influencing energy consumption.
3. Benchmarking: Comparing energy use data to industry standards, best practices, or historical performance to assess the efficiency of energy systems and equipment.
4. Energy audits: Conducting detailed assessments of energy use, including on-site inspections, equipment testing, and analysis of energy use patterns to identify opportunities for improvement.
5. Recommendations: Creating and ranking suggestions for enhancing energy efficiency, minimizing energy wastage, and maximizing energy utilization according to the results of the analysis.
6. Implementation: Implementing energy-saving measures, such as upgrading equipment, improving building insulation, optimizing HVAC systems, and implementing energy management strategies to reduce energy consumption and costs.

Organizations can enhance their understanding of energy use,

identify areas for improvement, and create plans to decrease energy consumption and costs, as well as minimize environmental impact, by analyzing energy use. Table 1 displays the energy inputs, outputs, and coefficients related to the production of soybean and peanut.

There are four primary energy indicators that measure energy consumption efficiency, energy productivity, specific energy, and net energy (Equations (2)–(5)) (Kaab et al., 2019). The energy consumption efficiency index shows how much energy has been harvested for each MJ ha^{-1} of energy consumed for production. The larger the ratio, the higher the energy efficiency. The energy productivity index also indicates that for every kg MJ^{-1} of input energy, how many kg of data are obtained. Also, the specific energy index is defined as the ratio of the total energy input to the performance of the product. Therefore, the larger the value of this index is, the more energy is wasted, and finally, the net energy index is the net energy output.

$$\text{Energy use efficiency} = \frac{\text{Output energy (MJ)}}{\text{Input energy (MJ)}} \tag{2}$$

$$\text{Energy productivity} = \frac{\text{Production (kg)}}{\text{Input energy (MJ)}} \tag{3}$$

$$\text{Specific energy} = \frac{\text{Input energy (MJ)}}{\text{Production (kg)}} \tag{4}$$

$$\text{Net energy} = \text{Output energy (MJ)} - \text{Input energy (MJ)} \tag{5}$$

2.3. LCA analysis

LCA is a technique utilized to assess the environmental effects of a product or process from its inception to its disposal. The objective of LCA is to pinpoint and measure the environmental consequences of a product or process, encompassing its carbon emissions, energy usage, water consumption, and waste production (Elyasi et al., 2022; Taherzadeh-Shalmaei et al., 2023). During the inventory analysis phase, data is collected on the inputs and outputs of the product or process, such as raw materials, energy use, emissions, and waste generation. The impact assessment phase involves evaluating the potential environmental impacts of these inputs and outputs, such as greenhouse gas emissions, water pollution, and resource depletion (Nabavi-Pelesaraei et al., 2019a; Hamidinasab et al., 2023). The interpretation of LCA analysis results is crucial in identifying improvement opportunities and guiding decision-making. Businesses, governments, and organizations commonly utilize LCA analysis to inform decisions regarding product design, manufacturing processes, and waste management strategies. Additionally, it can be used to compare the environmental performance of various products or processes and pinpoint areas for improvement in order to minimize environmental impacts (Kaab et al., 2020). Fig. 1 illustrates how the LCA stages define the parameters of soybean and peanut production systems.

The LCA steps including (Azizpanah et al., 2023; Hatim et al., 2023).

1. The LCA study aims to establish the goal and scope by defining the functional unit (FU), system boundaries, and impact categories to be assessed.
2. Collect data on the inputs and outputs of the product or process being studied, including raw materials, energy consumption, emissions, and waste generation.
3. Evaluate the environmental effects of the resources used and the waste produced by employing life cycle impact assessment tools like ReCiPe or similar methods.
4. Analyze the impact assessment results to pinpoint areas with the most significant impact and opportunities for enhancing the product or process.
5. Make recommendations for improving the environmental performance of the product or process, such as through changes in materials, production methods, or end-of-life management.
6. Communicate the results of the LCA study to stakeholders, such as consumers, regulators, and industry partners, to inform decision-making and promote sustainable practices.

Table 1
Energy inputs, outputs, and coefficients in the production of soybean and peanut.

Items	Unit	Energy equivalent (MJ $unit^{-1}$)	References
A. Inputs			
1. Human labor	h	1.96	Kaab et al. (2019)
2. Machinery	kg yr	62.7	Ordikhani et al. (2021)
3. Diesel fuel	L	56.31	Molaee Jafrodi et al. (2022)
4. Chemical fertilizers	kg		
(a) Nitrogen		66.14	Taherzadeh-Shalmaei et al. (2021)
(b) Phosphate (P_2O_5)		12.44	Khalaj et al. (2023)
(b) Potassium (K)		11.15	Kaab et al. (2023)
5. Farmyard manure	kg	0.3	Nabavi-Pelesaraei et al. (2019b)
6. Biocides	kg	120	Kaab et al. (2019)
7. Electricity	kWh	11.93	Mohammadi Kashka et al. (2023)
8. Soybean seed	kg	14.70	Mohammadi et al. (2013)
9. Peanut seed	kg	25.00	Nabavi-Pelesaraei et al. (2013)
B. Outputs	kg		
1. Soybean		14.70	Mohammadi et al. (2013)
2. Peanut		25.00	Nabavi-Pelesaraei et al. (2013)

2.3.1. Damage assessment of LCA

Damage assessment is the process of evaluating the extent of damage to buildings, infrastructure, and other property caused by natural disasters, accidents, or other events. This assessment is typically conducted by trained professionals such as engineers, architects, or insurance adjusters, who inspect the affected area and document the damage (Canaj et al., 2020; Maklavany et al., 2023). The assessment helps determine the cost of repairs, the safety of the structure, and the level of assistance needed for recovery efforts. It is an important step in the recovery process and can help guide decisions on rebuilding and resource allocation (Cheng et al., 2024). The incorporated human health, ecosystem quality, and resource scarcity as the three areas of protection. Characterization factors for endpoints, which are directly linked to these areas of protection, were calculated from midpoint characterization factors using a consistent mid-to-endpoint factor for each impact category. Have taken into account 17 midpoint impact categories (Fig. 2).

2.4. ICA optimization techniqh

The ICA is a metaheuristic optimization algorithm inspired by the concept of imperialism and competition among empires. It is used to solve complex optimization problems by modeling the process of imperialist expansion and competition for resources (Hadidi et al., 2013). The ICA algorithm is based on the principles of imperialism, where a set of candidate solutions (empires) compete with each other to become the dominant solution (imperialist) in the search space (Tahmasebi Boldaji et al., 2023). The algorithm consists of two main phases: the imperialistic competition phase and the assimilation phase. In the imperialistic competition phase, the empires compete with each other based on their fitness values, where the fitter empires dominate the weaker ones. The dominant empires expand their territories by

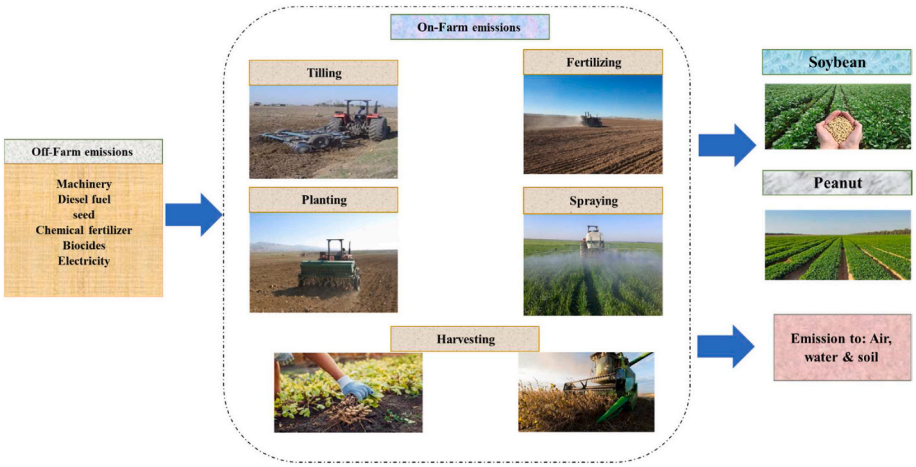

Fig. 1. The LCA stages delineate the boundaries of soybean and peanut production systems.

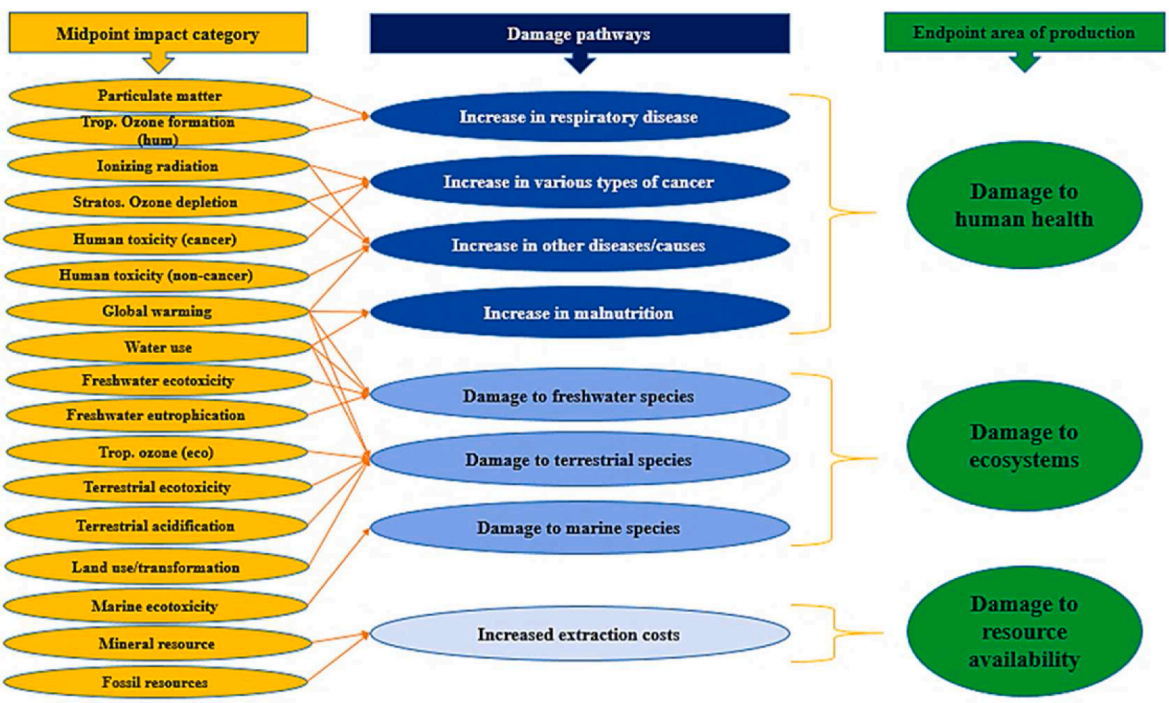

Fig. 2. The relationship between midpoints and endpoints is determined using the ReCiPe2016 method of LCA.

assimilating the weaker ones, leading to a redistribution of resources and a shift in the global solution space. In the assimilation phase, the dominant empires assimilate the weaker ones by exchanging information and resources, leading to the exploration and exploitation of the search space (i Canals et al., 2007). This process allows the algorithm to converge towards the optimal solution by balancing exploration and exploitation. The ICA algorithm has been successfully applied to various optimization problems, including engineering design, scheduling, and data mining. It is known for its ability to effectively handle complex, multimodal, and non-convex search spaces, making it a popular choice for solving real-world optimization problems. Overall, the Imperialist Competitive Algorithm is a powerful optimization technique that leverages the principles of imperialism and competition to efficiently search for optimal solutions in complex problem domains (Talaviya et al., 2020).

Multi-objective optimization involves considering two or more conflicting objectives simultaneously, without detailed information on their interactions. The research focuses on addressing mathematical optimization problems with multiple objectives. It aims to maximize energy output and minimize environmental emissions in soybean and peanut production based on input energy. The study employs an evolutionary optimization algorithm, inspired by imperialistic competition, known as the ICA, which uses a global search strategy to handle diverse optimization tasks (Hadidi et al., 2013). The ICA algorithm has shown to be very effective in terms of how quickly it converges and its ability to find the best possible solution, which makes it a good fit for this study. In ICA, the individuals in the population are called countries, and the initial population is made up of these countries. The algorithm represents the problem's solution space as a search space (Pardo et al., 2022). The initial distribution of colonies among imperialist countries is based on their power, leading to the formation of empires. Once the empires are established, a policy of assimilation is put in place for the colonies within each empire as they become more closely aligned with their respective imperialist country (Okokpujie et al., 2023). Empires engage

in competition to acquire and dominate colonies from other empires, leading to a weakening of smaller empires and a strengthening of larger ones (Boix-Cots et al., 2023). The total power is calculated using Eq. (6).

$$T \bullet C\bullet_n = Cost\ (Imperialist_n) + \vartheta\ mean\ \{\text{cost(colonies of empire}_n)\} \quad (6)$$

The total cost of the nth empire, $T \bullet C\bullet_n$, is determined by the positive number ϑ, which falls between zero and one and is typically assumed to be close to zero, around 0.1.

The basic iteration of ICA involves selecting weak colonies from weak empires and initiating a competition for them among all empires. The assimilation policy and imperialist competition are the most important operators in ICA, and the algorithm's flowchart can be found in Fig. 3 (Baldassarri et al., 2017).

3. Results and discussion

3.1. Inputs-outputs energy analysis

By calculating the average of the collected samples, the energy consumption of inputs was determined. Table 2 presents the energy input and output for soybean and peanut production, with each input evaluated based on its FU. A comparison of the energy consumption of different production systems was made. The total energy consumption of inputs for soybean production is 42748.10 MJ ha^{-1}, while the output energy is 40417.28 MJ ha^{-1}. Similarly, the total energy consumption of inputs for peanut production is 28677.36 MJ ha^{-1}, with the output energy also being 28677.36 MJ ha^{-1}. Shiv Kumar Lohan et al. (2023) the energy productivity of five major vegetable crops (potato, tomato, muskmelon, garden pea, cauliflower) production systems was investigated. The study revealed that potato cultivation had the highest energy consumption at 53,412 ± 2388 MJ ha^{-1}, followed by tomato at 47,489 ± 1183 MJ ha^{-1}, cauliflower at 39,367 ± 1127 MJ ha^{-1}, muskmelon at 37,827 ± 856 MJ ha^{-1}, and garden pea at 24,625 ± 497 MJ ha^{-1}. The study also identified irrigation as the primary energy-consuming farm operation, followed by the transportation of farmyard manure, fertilizers, and produce. Ahmadbeyki et al. (2023) discovered that enhancing ecological conditions and managing various factors in greenhouse crops is associated with increased energy usage. The study indicated that energy consumption ranged from 405405.75 to 412911.99 MJ ha^{-1}, with diesel fuel accounting for over 60% of the total energy consumption. Furthermore, the energy output for cucumber and tomato was found to be 104982.94 and 228281.37 MJ ha^{-1}, respectively.

Fig. 4 illustrates the comparison of input consumption percentages and irrigation methods for soybean and peanut production. Nitrogen fertilizers play a crucial role in enhancing crop yield, necessitating a significant energy input of over 40%. The timing of nitrogen fertilizer application stands as a key management approach to boost nitrogen efficiency. Understanding the plants' needs during the growing season is essential for optimizing nitrogen consumption. Following nitrogen fertilizers, diesel fuel accounts for 24.87% and 24.85% of input consumption in soybean and peanut production, respectively. Boroun et al. (2023) utilized of ICA to examine energy consumption and the life cycle of sugar production in Plant (PS) and Ratoon (RS) sugarcane farms. The study revealed that diesel fuel inputs and chemical fertilizers were the least efficient energy consumers.

Table 3 compares the energy indices for soybean and peanut production systems. The data suggests that peanut production has a higher energy use efficiency compared to soybean production, indicating that soybean provides more energy to consumers. Additionally, peanut production shows higher energy productivity, requiring less energy per kg of crop compared to soybean cultivation. However, the specific energy results show the opposite of energy productivity. Furthermore, peanut cultivation has achieved a net energy value of 15024.17 MJ ha^{-1}, indicating a high level of net energy. Ahmadbeyki et al. (2023) reported that tomato had a higher energy use efficiency (0.55) compared to cucumber (0.26).

Table 2
Amounts of inputs-outputs energy in soybean and peanut production systems.

Items	Soybean		Peanut	
	Unit per ha	Energy use (MJ ha^{-1})	Unit per ha	Energy use (MJ ha^{-1})
1. Human labor (h)	76.57	150.08	693.12	1358.52
2. Machinery (h)	24.00	1504.80	16.52	1036.11
3. Diesel fuel (L)	190.67	10736.90	126.55	7126.03
4. Chemical fertilizers (kg)				
(a) Nitrogen	280.52	18553.92	174.62	11549.69
(b) Phosphate (P_2O_5)	110.70	1377.10	40.95	509.41
(c) Potassium (K)	–	–	65.40	729.21
6. Biocides (kg)	9.40	1128.00	4.05	
7. Electricity (kwh)	605.95	7228.98	345.87	4149.86
8. Seed (kg)	140.70	2068.29	69.30	1732.50
Total energy use (MJ)	**-**	**42748.10**	**-**	**28677.36**
B. Output (kg)				
1. Soybean	2749.48	40417.28	–	–
2. Peanut	–	–	3601.5	43701.54

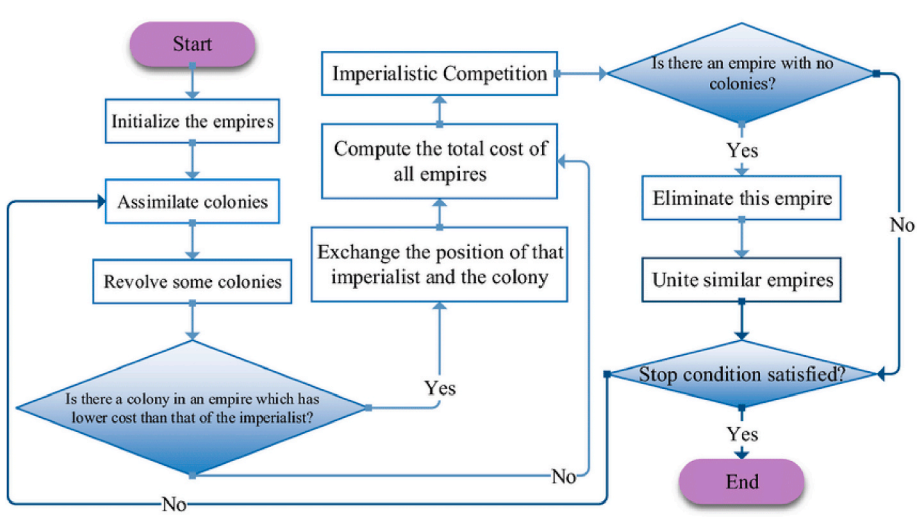

Fig. 3. Flowchart of imperialistic competition algorithm.

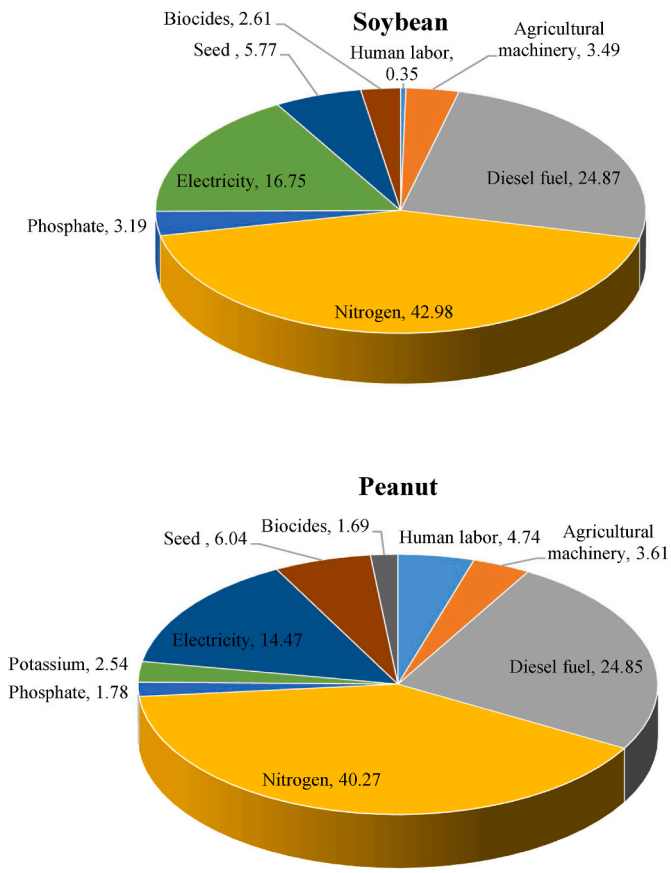

Fig. 4. The impact of energy sources on soybean and peanut production.

Table 3
Energy indices in soybean and peanut production systems.

Items	Soybean	Peanut
Energy use efficiency (ratio)	0.94	1.52
Energy productivity (kg MJ^{-1})	0.06	0.12
Specific energy (MJ kg^{-1})	15.75	7.96
Net energy gain (MJ ha^{-1})	−2752.91	15024.17

3.2. LCA analysis

Collecting and analyzing data is a critical step in the LCA process as it helps to identify the inputs and outputs of a product's life cycle. It is important to ensure that data collection is accurate in order to prevent any issues that may arise from a lack of data. When gathering data from public sources, it is important to maintain transparency and accuracy by clearly identifying reference sources. The EcoInvent database is often used as a primary resource. The collected data should be applicable to the specific FU defined in the objectives of the life cycle assessment (LCA). Inventory flow can be classified according to the scope of the system under analysis. Calculating environmental emissions from input production can be done using Off-Farm emissions. It is crucial to take into account all activities within the system boundary, including both upstream and downstream processes. It is also important to assess the quality of the data and address any uncertainties to ensure the reliability of the results. The results in Table 4 show that the production inputs for soybean and peanut production systems lead to on-farm emissions. Diesel fuel and chemical fertilizers are the primary contributors to these emissions. Soybeans have higher levels of diesel fuel pollutants due to reduced usage, while peanuts have a higher prevalence of contaminants associated with diesel fuel.

Table 5 presents the damage assessment results for various soybean and peanut production. It is evident that resources have the most significant environmental impact, with higher greenhouse gas emissions associated with soybean compared to peanut. The level of ecosystem impact in soybean and peanut production is 8.49E-05 and 3.98E-05, respectively. Mostashari-Rad et al. (2021) a study was carried out using the ReCiPe 2016 method to evaluate the environmental impact of horticultural products. The findings indicated that citrus, hazelnut, kiwifruit, tea, and watermelon had a greater impact on the environment and human health in comparison to other categories in terms of resource usage. Furthermore, Litskas et al. (2017), Bosco et al. (2011), and (Point et al. (2012) reported greenhouse gas emissions of 0.155 kg CO_{2eq}, 0.15–0.3 kg CO_{2eq}, and 0.8 kg CO_{2eq}, respectively. Ahmadbeyki et al. (2023) the LCA results emphasized that carbon dioxide emissions were the most prominent among 18 air pollutants, mainly due to diesel fuel usage. The assessment of harm, carried out using the ReCiPe 2016 method, showed human health impacts of 0.012 and 0.004 DALY for cucumber and tomato, respectively. Importantly, emissions occurring

Table 4
On-Farm emissions in soybean and peanut production systems based on 1 ha.

	Soybean	Peanut
1. Emissions by diesel fuel to air (kg)		
(a). Carbon dioxide (CO_2)	799.89	530.88
(b). Sulfur dioxide (SO_2)	0.25	0.17
(c). Methane (CH_4)	0.03	0.021
(d). Benzene	0.001	0.001
(e). Cadmium (Cd)	0.000003	0.000001
(f). Chromium (Cr)	0.00001	0.000008
(g). Copper (Cu)	0.0004	0.0002
(h). Dinitrogen monoxide (N_2O)	0.03	0.02
(i). Nickel (Ni)	0.00001	0.00001
(j). Zink (Zn)	0.0002	0.0001
(k). Benzo (a) pyrene	0.000008	0.000005
(l). Ammonia (NH_3)	0.005	0.003
(m). Selenium (Se)	0.000003	0.000001
(n). PAH (polycyclic hydrocarbons)	0.0008	0.0005
(o). Hydro carbons (HC, as NMVOC)	0.73	0.48
(p). Nitrogen oxides (NOx)	11.38	7.55
(q). Carbon monoxide (CO)	1.61	1.06
(r). Particulates (b2.5 μm)	1.148	0.76
2. Emissions by fertilizers to air (kg)		
(a). Ammonia (NH3) by chemical fertilizers	34.06	21.20
3. Emissions by fertilizers to water (kg)		
(a). Nitrate	37.26	23.20
(b). Phosphate	2.41	0.89
4. Emission by N_2O of fertilizers and soil to air (kg)		
(a). Nitrogen oxides (NOx)	7.153	4.45
5. Emission by human labor to air (kg)		
(a). Carbon dioxide (CO_2)	53.60	485.18
6. Emission by heavy metals of fertilizers to soil (mg)		
(a). Cadmium (Cd)	11701.50	4766.80
(b). Copper (Cu)	30208.55	13585.88
(c). Zink (Zn)	269822.675	114934.74
(d). Lead (Pb)	1534407.525	950951.02
(e). Nickel (Ni)	28224.3725	12215.86
(f). Chromium (Cr)	159674.3975	65272.73
(g). Mercury (Hg)	105.5425	52.66

Table 5
Values of the damage assessment per one in soybean and peanut production systems.

Items	Unit	Soybean	Peanut
Human health	DALY [a]	0.08	0.03
Ecosystems	species.yr [b]	8.49E-05	3.98E-05
Resources	USD2013	131.13	64.48

[a] DALY: disability adjusted life years. A damage of 1 is equal to: loss of 1 life year of 1 individual, or 1 person suffers 4 years from a disability with a weight of 0.25.
[b] Species.yr: the unit for ecosystems is the local species loss integrated over time.

on-site had the greatest influence on human health, accounting for 82% and 78% for cucumber and tomato, respectively. Boroun et al. (2023), it was discovered that RS farms generally had lower environmental burdens in most production areas compared to PS farms. However, the excessive use of chemical fertilizers on RS farms resulted in significant aquatic ecotoxicity and eutrophication. Despite consuming more energy, PS farms were found to be more efficient. The study suggests that to mitigate environmental impacts in the sugar industry, the most effective strategy would be to focus on reducing aquatic ecotoxicity and eutrophication.

Fig. 5 depicts the emissions distribution from different sources in the soybean and peanut production. Emissions from both production methods have a notable impact on human health, with soybean responsible for 70% of emissions and peanut contributing 65%. Diesel fuel and nitrogen fertilizer are the inputs with the most significant environmental impact, making up over 50% of the overall impact. Effective nitrogen fertilizer management is essential for maximizing crop growth and yield, making it a top priority for researchers and farmers. Many areas globally have imposed stringent rules on chemical fertilizer usage in farming to avoid excessive amounts of these substances from polluting the environment. This not only safeguards the environment and public health but also delivers economic advantages like lower costs, enhanced productivity, and resource preservation. Steenwerth et al. (2015) two methods for sustainable agricultural production, mineral fertilizer and compost fertilizer, have been proposed as effective fertilizer management techniques. Careful consideration of the suitable application of fertilizers is crucial to minimize environmental impacts and ensure efficient farming practices.

3.3. Optimization results

Table 6 displays the findings from an analysis of ideal energy utilization and energy conservation in soybean and peanut production systems. The research employed ICA to calculate the total optimized energy consumption, resulting in 36672.77 MJ ha^{-1} for soybean production and 23485.45 MJ ha^{-1} for peanut production. Furthermore, energy savings of 6075.33 and 5191.91 MJ ha^{-1} were realized for soybean and peanut production, respectively. Table 7 displays the environmental impact category quantities for ICA analysis of soybean and peanut production. The table indicates that the resources per ton for soybean production amounted to 111.25 USD 2013, while for peanut production, the resources per ton for energy use were calculated to be 45.43 USD 2013. Khanali et al. (2021) a research study was carried out to explore multi-objective optimization of energy usage and environmental emissions in walnut production using the ICA. The results revealed a substantial decrease in total energy consumption by 19,316 MJ ha^{-1} (approximately 62%), with gasoline exhibiting the most noteworthy energy savings at 58% compared to other inputs. Furthermore, the overall weighted emissions were reduced by approximately 1.47 Pt (about 40%). These findings suggest that timely maintenance can greatly aid orchardists in achieving near-optimal conditions. Moreover, the implementation of the ICA not only offers an efficient model for walnut production but also has the potential for global application across various crops. The energy use efficiency of fattening farms was assessed using data envelopment analysis (DEA), which identified both efficient and inefficient operations and measured energy wastage. The study also investigated how optimizing energy usage could impact

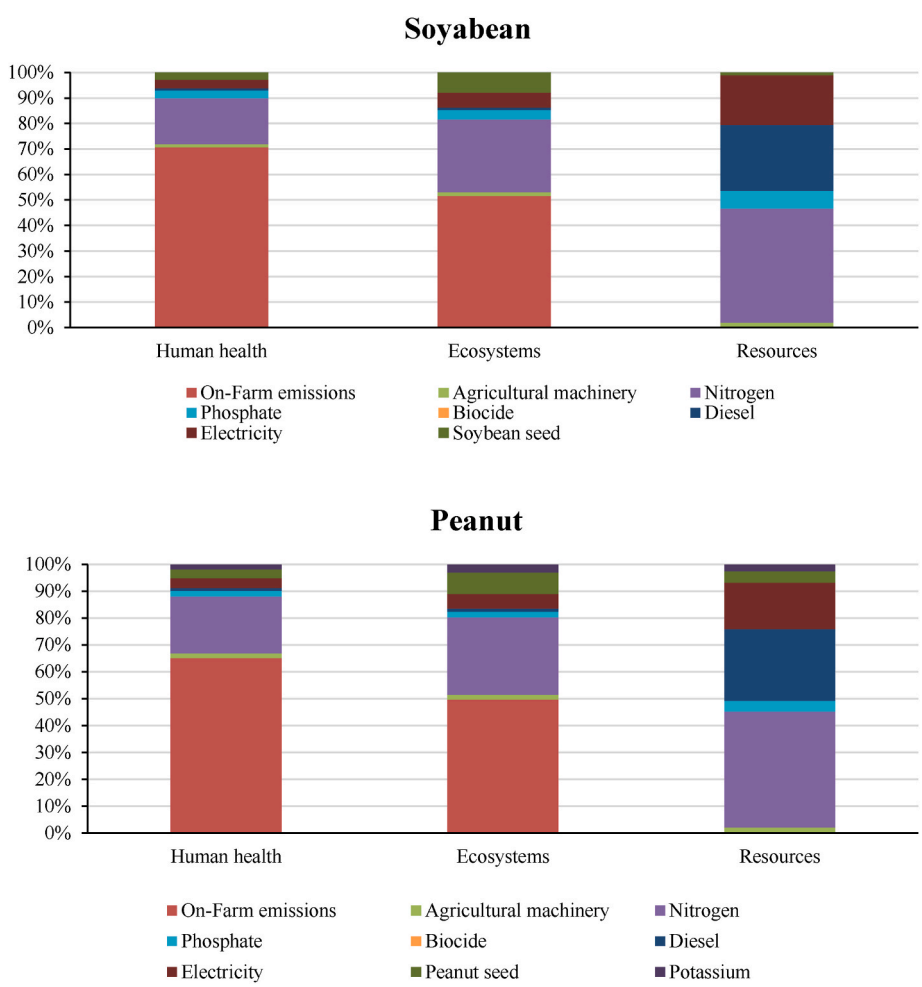

Fig. 5. Contribution of different inputs in the damages categories for soybean and peanut production systems.

Table 6
Optimum energy requirement and saving energy for soybean and peanut production systems based on ICA techniqh.

Inputs	Optimum energy requirement (MJ ha^{-1})		Saving energy (MJ ha^{-1})		Saving energy (%)	
	Soybean	Peanut	Soybean	Peanut	Soybean	Peanut
1. Human labor	120.17	1184.69	29.91	173.83	19.92	12.79
2. Machinery	1321.84	885.63	182.96	150.48	12.15	14.52
3. Diesel fuel	10151.70	5469.23	585.20	1656.80	5.45	23.24
4.Nitrogen	15694.39	9456.32	2859.53	2093.37	15.41	18.12
5.Phosphate	1125.31	441.23	251.79	68.18	18.28	13.38
6. Potassium	–	554.21	–	175.00		23.99
6. Biocides	945.75	354.69	182.25	131.31	16.15	2.73
7. Electricity	5412.98	3589.87	1816.00	559.99	25.12	13.49
8. Seed	1900.63	1549.58	167.66	182.92	8.10	10.55
Total energy input	**36672.77**	**23485.45**	**6075.33**	**5191.91**	**14.51**	**18.10**

Table 7
Optimum values of the environmental impacts of the soybean and peanut production systems per tonne based on ICA techniqh.

Impact category	Units	Soybean	Peanut
Human health	DALY [a]	0.06	0.02
Ecosystems	species.yr [b]	6.51E-05	2.81E-05
Resources	USD2013	111.25	45.43

greenhouse gas emissions, comparing emissions from current and optimal energy consumption. Results indicated that 43.33% of fattening farms operated efficiently, while 56.67% were inefficient based on a constant returns to scale model. The average technical, pure technical, and scale efficiencies of the farms were determined to be 0.937, 0.953, and 0.983, respectively. The study found that total energy consumption was 24,003 (MJ $calf^{-1}$), with an optimal requirement of 21,931 (MJ $calf^{-1}$), resulting in an energy saving target ratio of 8.63%. Feed intake contributed to 53% of total energy savings, followed by fossil fuels at 31%. The total greenhouse gas emissions were assessed at 1176 (kg $CO_{2\,eq}$. $calf^{-1}$ 6 $months^{-1}$), which could potentially be reduced to 1073 (kg CO_{2eq}. $calf^{-1}$ 6 $months^{-1}$) through optimal energy consumption (Hosseinzadeh-Bandbafha et al., 2017). Energy consumption and its environmental impacts have been significant topics in recent centuries. Agricultural systems serve as both energy users and suppliers in the form of bioenergy, playing a crucial role in global economics and food security. This sector utilizes a substantial amount of energy from various sources, and research into energy flow in crop production, particularly in developing countries, has revealed significant inefficiencies. To distinguish between efficient and inefficient farms, a clustering model based on the ICA was developed. Watermelon farms were surveyed and clustered based on three features: greenhouse gas emissions, input energy, and farm size. The results indicated that the best cluster outperformed the other two clusters by 20–46% in energy usage and 22–52% in CO_2 emissions. The average total energy input and GHG emissions for the best cluster were calculated as 43,423 MJ per hectare and 8120 CO_{2eq}. This study demonstrates the successful application of ICA for optimizing energy use in cropping systems, leading to improved environmental and energy performance (Khoshnevisan et al., 2015).

4. Conclusions

The objective is to enhance energy use efficiency in agriculture, mitigate environmental impacts such as greenhouse gas emissions, and assess factors such as carbon footprint, water usage, and soil quality. The research reveals that soybean production consumes 43170.20 MJ ha^{-1} of total energy and yields 40417.28 MJ ha^{-1} of output energy, while peanut production requires 28677.36 MJ ha^{-1} of total energy and produces 28677.36 MJ ha^{-1} of output energy. The LCA results show that on-farm emissions are associated with production inputs for both soybean and peanut systems, with diesel fuel and chemical fertilizers being the primary contributors. Soybean farming exhibits higher levels of diesel fuel pollutants due to reduced usage, whereas peanuts have a higher prevalence of contaminants associated with diesel fuel. ICA was utilized to calculate the optimized energy consumption, resulting in 36672.77 MJ ha^{-1} for soybean production and 23485.45 MJ ha^{-1} for peanut production, leading to energy savings of 6075.33 MJ ha^{-1} and 5191.91 MJ ha^{-1}, respectively. Additionally, the environmental impact category quantities for ICA analysis of soybean and peanut production reveal the resources per ton for soybean production to be 111.25 USD2013 and for peanut production to be 45.43 USD2013 in terms of energy use. The study aims to identify sustainable farming methods that minimize environmental effects while maximizing crop yield and ultimately support the adoption of sustainable farming practices in the area. It suggests that ICA can help farmers reduce energy consumption and minimize environmental impact, leading to more sustainable and environmentally friendly farming practices. The study underscores the potential for further research and implementation of ICA in farming practices to significantly improve energy use efficiency and environmental sustainability in the soybean and peanut farming industry. Overall, the study emphasizes the potential benefits of using ICA to optimize resource allocation and decision-making processes in farming, ultimately leading to more sustainable and environmentally friendly practices in the agricultural sector.

CRediT authorship contribution statement

Abbas Sharifi Nia: Writing – original draft, Methodology, Investigation, Formal analysis, Data curation. **Mohammad Gholami Parashkoohi:** Writing – original draft, Supervision, Software, Resources. **Davood Mohammad Zamani:** Software, Resources, Methodology, Formal analysis. **Hamed Afshari:** Validation, Software, Methodology, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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